1. Executive Summary & Core Concept

RAZIN is an Autonomous Robotic Forensic Evidence Investigation System designed to assist forensic teams by using Artificial Intelligence (AI) and Computer Vision (CV). The primary goal of the project's first phase is to build an autonomous or semi-autonomous robotic system capable of conducting preliminary surveys of crime scenes and documenting evidence. By utilizing a mobile robot, the system aims to automate processes that usually rely heavily on human observation, such as photographing the scene, locating evidence, and recording data.

2. System Architecture & Modularity

RAZIN is built with a highly modular architecture to ensure the project is scalable and easy to upgrade. Instead of being developed as a single massive software block, it is divided into independent modules.

- **Key Modules:** The architecture includes Simulation, Robot Control, Sensors, Navigation, Computer Vision, Evidence Management, Localization, and a user-facing Dashboard.
- **Modularity Advantage:** This design allows developers to test each phase independently and easily swap out technologies in the future (e.g., upgrading a standard camera to a depth sensor or moving from basic detection to advanced image segmentation) without rebuilding the entire system.

3. The Minimum Viable Product (MVP) - Phase 1

For the initial 10-day development sprint, the team is focusing strictly on an MVP (Phase 1). The MVP will operate entirely within a simulated environment to avoid the immediate need for physical hardware.

- **Simulation Environment:** The robot will be simulated using the **Webots** platform, which provides the necessary physics, environments (starting with a single room), and sensor simulations.
- **Target Evidence:** To maintain a realistic scope, the AI will initially be trained to detect only three classes of evidence: Bloodstains, Footprints/Shoeprints, and Digital Devices (such as smartphones and laptops).
- **Operational Flow:** The simulated robot will navigate through predefined waypoints. As it moves, its camera captures frames that are fed into the AI model. When evidence is detected, the system generates an Evidence ID, captures an image, records the timestamp and the robot's exact coordinates, and stores this data in a SQLite database to update a Digital Evidence Map.

4. Technical AI Approach

To optimize resources and time, the project will use **Transfer Learning** rather than building a deep learning model from scratch.

- The team will utilize a pretrained **YOLO (You Only Look Once)** object detection model.
- This model will be fine-tuned using a specific forensic dataset, which drastically reduces training time, requires less computational power, and minimizes the amount of data needed.

5. Project Management & Team Structure

The two-person development team has divided the workload to build the MVP efficiently in 10 days.

- **Member 1 (Robotics & Simulation):** Responsible for the Webots environment, robot movement, navigation, and camera setup.
- **Member 2 (AI & Forensics):** Responsible for dataset preparation, YOLO fine-tuning, computer vision, evidence management, and the database.
- **Collaboration:** Both members will work together on system integration, testing, and building the Streamlit Dashboard, which will display a live camera feed, an evidence list, a scene map, and the robot's status.
- **Development Rule:** A strict sequence must be followed: simulation must work before AI is integrated, and data must be gathered before the dashboard is built.

6. Long-Term Vision & Roadmap

RAZIN is structured for continuous evolution beyond the MVP, with a multi-phase roadmap:

- **Phase 2 & 3:** Upgrading to advanced perception (depth, object tracking) and fully autonomous navigation using SLAM (Simultaneous Localization and Mapping).
- **Phase 4 & 5:** Implementing advanced forensic analysis (e.g., footprint similarity matching) and multi-sensor fusion (combining RGB, LiDAR, Thermal, and IMU data).
- **Phase 6 & 7:** Adding a Large Language Model (LLM) layer to serve as an Intelligent Investigation Assistant that can summarize scenes, eventually culminating in a fully integrated, physical robotic platform.

Ultimately, the goal of the initial sprint is not to build a physical robot, but to deliver a stable, versioned digital foundation (RAZIN v0.1) that successfully proves the concept of autonomous forensic surveying.